

SAMUEL ELÍAS ORTEGA RODRÍGUEZ

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EDUCATION

Universidad del Norte

Expected 2029 (4th Sem | Started 2025)

Candidate for Bachelor of Science in Systems Engineering (ABET-Accredited Curriculum)
International Degree Equivalent: B.S. in Computer Science & Engineering | Barranquilla, Colombia

- Relevant Coursework:** Object-Oriented Programming (Java), Data Structures & Algorithms, Discrete Mathematics.

TECHNICAL SKILLS

Languages: Java (OOP, Collections, Exception Handling), Python (ETL Pipelines, Automation, File I/O), SQL (PostgreSQL, SQLite), Bash/PowerShell.

Technologies & Libraries: REST APIs (Socrata SODA), Pandas, Requests, BeautifulSoup, OpenPyXL, JSON-LD / Schema.org.

Developer Tools & Infrastructure: Git, GitHub (Feature Branching, PRs, Code Reviews), Cloudflare Pages, VS Code, CI/CD Automation Scripts.

Core Concepts: Data Structures & Algorithms, Object-Oriented Design, Big-O Complexity Analysis, Data Validation, Defensive Programming.

Languages Proficiency: English (B2 – Professional Working Proficiency, Technical Documentation), Spanish (Native).

TECHNICAL PROJECTS

Automated Municipal Data Ingestion & Analytics Pipeline

2026

Python, SODA REST API, Cloudflare, ETL, OpenPyXL

- Architected** an automated end-to-end Python ETL pipeline ingesting municipal commercial building filings from Chicago's public SODA 2.0 REST endpoint, sustaining zero-failure execution at \$0.00/month cloud infrastructure cost.
- Engineered** server-side SoQL filters and schema normalization modules to extract commercial CapEx projects ($\geq \$50k$), pruning redundant payload attributes and enforcing 100% PII-free data privacy compliance.
- Automated** recurring delivery of weekly multi-tab Excel workbooks (.xlsx) using OpenPyXL with standardized accounting formatting (\$#,##0) and generated executive PDF digests via scheduled headless scripts.
- Deployed** edge web distribution platform on Cloudflare Pages integrating Schema.org JSON-LD structured metadata (DataCatalog, Dataset, Product), passing 100% of automated pre-flight SEO and validation audits.

Real-Time Interactive Game Application

2025 – 2026

Java, OOP, Design Patterns, Git, GitHub — GuayabaLabs GitHub Organization

- Architected** a modular desktop game engine prototype in Java, decoupling scene management, physics transitions, and event dispatching through rigorous implementation of the State and Observer design patterns.
- Engineered** a deterministic game loop prototype with real-time delta-time calculations, targeting 60 FPS update-render synchronization while stabilizing memory through object lifecycle optimization.
- Constructed** boundary collision-detection algorithms and dynamic entity management systems utilizing Java Collections (ArrayList, HashMap) for sub-millisecond runtime entity tracking.
- Coordinated** distributed engineering workflows within the GuayabaLabs GitHub team, enforcing code quality through feature branching, strict PR reviews, and comprehensive technical Markdown specifications.

Algorithmic Data Validation & Optimization Utility

2025

Python, Hash Tables, Algorithms, Big-O Analysis

- Optimized** structured dataset processing efficiency from $O(N^2)$ quadratic complexity to $O(N)$ linear-time performance by eliminating nested loops and refactoring lookups to hash-based indexing and set operations.
- Engineered** defensive exception-handling routines and schema validation rules to isolate boundary failures, empty rows, and malformed inputs, securing 100% script execution uptime across edge-case benchmarks.
- Authored** a comprehensive test suite incorporating boundary values, extreme strings, and corrupt delimiters to systematically trace execution paths, validate deterministic error recovery, and assert data sanity.